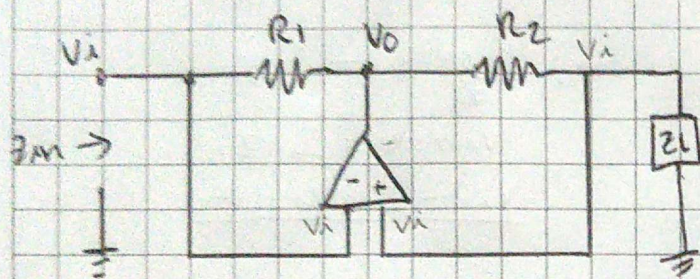


EJERCICIO 9



$$Z_{in} = \frac{V_{in}}{I_{in}} \quad \text{con} \quad I_{in} = \frac{V_i - V_o}{R_1} \quad (I)$$

$$\frac{V_o - V_i}{R_2} = \frac{V_i - 0}{Z_L} \quad (II)$$

En función de la carga conectada:

$$V_o - V_i = \frac{V_i R_2}{Z_L}$$

$$V_o = V_i \left(\frac{R_2}{Z_L} + 1 \right)$$

$$I_{in} = \frac{V_i}{R_1} - \frac{V_i}{R_1} \left(\frac{R_2}{Z_L} + 1 \right) \rightarrow \frac{R_2 + Z_L}{Z_L}$$

$$I_{in} = V_i \left(\frac{1}{R_1} - \frac{R_2 + Z_L}{R_1 Z_L} \right) = V_i \frac{Z_L - (R_2 + Z_L)}{R_1 Z_L}$$

$$I_{in} = V_i \cdot \left(\frac{-R_2}{Z_L R_1} \right)$$

$$Z_{in} = \frac{V_{in}}{I_{in}} = \frac{-Z_L R_1}{R_2}$$